STPM 2014 PHYSICS 960/2

Questions and Answers

Values of constants: $e = 1.60 \times 10^{-19}\ C$, $m_p = 1.67 \times 10^{-27}\ kg$, $\varepsilon_0 = 8.85 \times 10^{-12}\ F\ m^{-1}$, $k = \frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^9\ N\ m^2\ C^{-2}$, $\mu_0 = 4\pi \times 10^{-7}\ H\ m^{-1}$.

Section A [15 marks]

1. Two point charges +35 μC and Q are placed 300 cm apart. If they attract each other with a force of 0.525 N, what is the charge of Q?

A +167 μC B +15 μC C −15 μC D −167 μC

Answer: C

$$|Q| = \frac{Fr^2}{kq} = \frac{(0.525)(3.00)^2}{(9.0 \times 10^9)(35 \times 10^{-6})} = 1.5 \times 10^{-5}\ C$$

They attract, so Q is negative: −15 μC.

Explanation: 300 cm = 3.00 m.

2. A dust particle of mass 2.0×10^{-18} kg carries a charge of -1.60×10^{-19} C. It is placed in a uniform downward electric field strength of 200 N C^{-1} and a uniform gravitational field strength of 9.81 N kg^{-1}. What is the magnitude and direction of the resultant force of the dust particle?

A 1.24×10^{-17} N, upwards B 1.24×10^{-17} N, downwards

C 5.16×10^{-17} N, upwards D 5.16×10^{-17} N, downwards

Answer: A

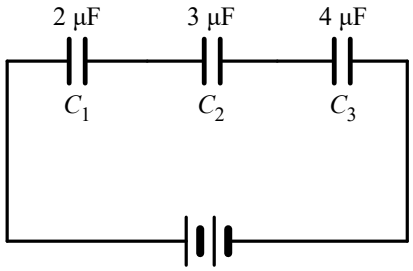
Weight = $mg = (2.0 \times 10^{-18})(9.81) = 1.96 \times 10^{-17}$ N downwards

Electric force = $qE = (1.60 \times 10^{-19})(200) = 3.20 \times 10^{-17}$ N upwards (negative charge in a downward field)

Resultant = $3.20 - 1.96 = 1.24 \times 10^{-17}$ N, upwards.

Explanation: A negative charge feels a force opposite to the field.

3. Three capacitors C_1, C_2 and C_3 are connected in series to a battery as shown in the diagram below.



The capacitances of each capacitor are 2 μF, 3 μF and 4 μF respectively. Which statement about the circuit is true?

A The equivalent capacitance is 9 μF.
B The charge in capacitor C_3 is the biggest.
C The potential difference across C_1 is the biggest.
D The energy stored in capacitor C_3 is greater than that of capacitor C_2.

Answer: C

In series every capacitor has the same charge; $V = \frac{Q}{C}$ is largest for the smallest capacitance, C_1

(C)

Explanation: The equivalent capacitance is 0.92 μF; C_3 stores the least energy.

4. A parallel plate capacitor in a vacuum is charged, and it stores an energy U. If the capacitor is then filled with a medium of relative permittivity ε_r and it is charged with the same potential, what is the energy stored in the capacitor?

A $\varepsilon_r U$ B $\varepsilon_r \varepsilon_0 U$ C $\frac{\varepsilon_0}{\varepsilon_r} U$ D $\frac{\varepsilon_r}{\varepsilon_0} U$

Answer: A

Same p.d.: $U = \frac{1}{2} CV^2 \propto C$, and C becomes $\varepsilon_r C$, so the energy becomes $\varepsilon_r U$

Explanation: With a fixed charge instead, the energy would fall to U/ε_r.

5. The drift velocity of free electrons in a conductor is very small, but a bulb can be lit up almost immediately after the switch is closed. Which statement explains this situation?

A The mass of an electron is very small.
B The random speed of free electrons is high.
C Free electrons move with the speed of light.
D All free electrons drift simultaneously.

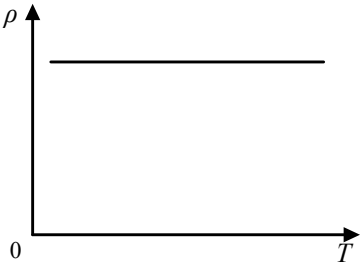
Answer: D

The wire is full of free electrons; when the switch is closed the field is set up almost instantly and all the free electrons start drifting at the same time (D).

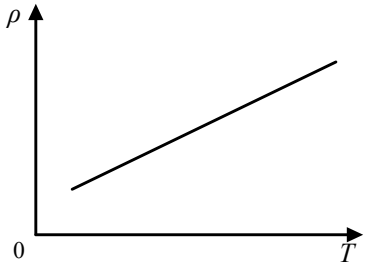
Explanation: Electrons already in the filament carry the current immediately.

6. Which graph shows the variation of resistivity ρ with temperature T of a superconductor?

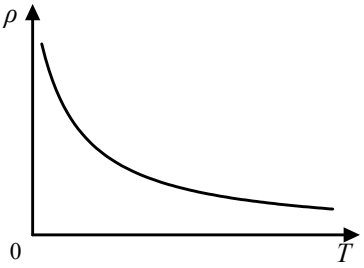
A



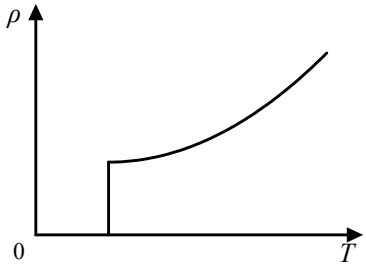
B



C



D



Answer: D

A superconductor's resistivity drops suddenly to zero below its critical temperature (D).

Explanation: Above the critical temperature it behaves like a normal metal.

7. A bulb dissipates a power of 60 W when connected to a power supply of 240 V. If the bulb is then connected to a power supply of 110 V, what is the current flowing through the bulb, and the power dissipated by the bulb?

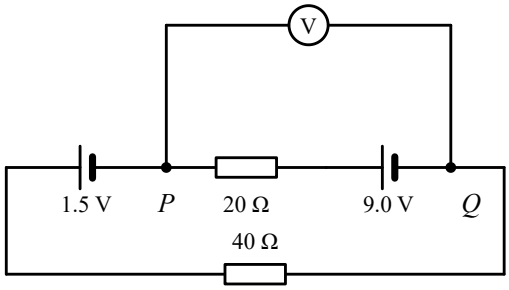
	Current	*Power*
A	0.11 A	12.6 W
B	0.11 A	27.5 W
C	0.25 A	27.5 W
D	0.25 A	60.0 W

Answer: A

$$R = \frac{240^2}{60} = 960\ \Omega;\ I = \frac{110}{960} = 0.11\ \text{A};\ P = \frac{110^2}{960} = 12.6\ \text{W}$$

Explanation: Assume the filament resistance stays constant.

8. A circuit consisting of two cells of e.m.f. 1.5 V and 9.0 V connected in series with two resistors 20 Ω and 40 Ω is shown in the diagram below.



A voltmeter is connected across points P and Q. What is the reading of the voltmeter?

A 1.0 V B 5.0 V C 6.5 V D 7.5 V

Answer: C

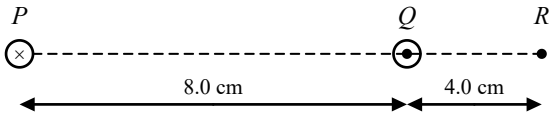
Taking the polarities as drawn, the 9.0 V and 1.5 V cells oppose each other: net e.m.f. = 7.5 V.

$$I = \frac{7.5}{20 + 40} = 0.125 \text{ A}$$

$$V_{PQ} = 9.0 - (0.125)(20) = 6.5 \text{ V (check: } (0.125)(40) + 1.5 = 6.5 \text{ V)}$$

Explanation: The voltmeter reads the p.d. across the 20 Ω resistor and the 9.0 V cell together.

9. Two parallel conductors P and Q carrying equal currents of 2.0 A in the opposite directions are shown in the diagram below.



What is the magnetic flux density at point R?

A 4.2×10^{-5} T B 2.1×10^{-5} T C 1.3×10^{-5} T D 6.7×10^{-6} T

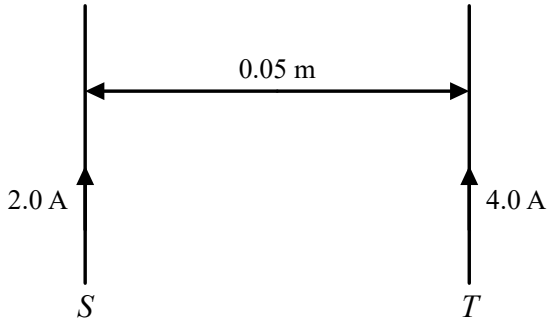
Answer: D

From P (12.0 cm): $\frac{\mu_0(2.0)}{2\pi(0.120)} = 3.3 \times 10^{-6}$ T; from Q (4.0 cm): $\frac{\mu_0(2.0)}{2\pi(0.040)} = 1.0 \times 10^{-5}$ T

The currents are opposite, so at R (outside both) the fields are opposite: $B = 1.0 \times 10^{-5} - 3.3 \times 10^{-6} = 6.7 \times 10^{-6}$ T.

Explanation: Use the right-hand grip rule for each wire.

10. Two long wires S and T which are separated by a distance of 0.05 m in a vacuum, and carrying currents of 2.0 A and 4.0 A respectively in the same direction are shown in the diagram below.



If S is fixed but T is free to move, what is the force per unit length and its direction that must be applied to T to prevent it from moving?

A 3.2×10^{-5} N m^{-1}, towards S
B 3.2×10^{-5} N m^{-1}, away from S
C 1.0×10^{-4} N m^{-1}, towards S
D 1.0×10^{-4} N m^{-1}, away from S

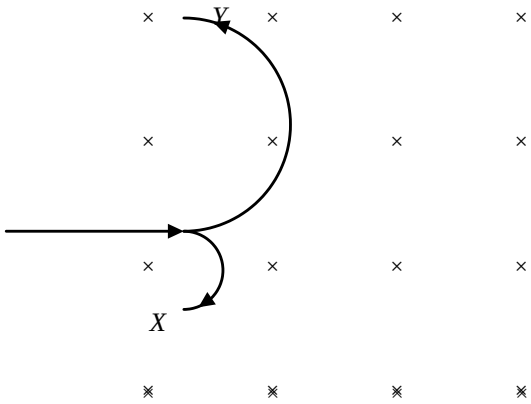
Answer: B

$$\frac{F}{l} = \frac{\mu_0(2.0)(4.0)}{2\pi(0.05)} = 3.2 \times 10^{-5} \text{ N m}^{-1}, \text{ attraction towards } S$$

To hold T still, an equal force must be applied away from S.

Explanation: Currents in the same direction attract.

11. The path of two charged particles X and Y which enter a uniform magnetic field perpendicularly with the same velocity are shown in the diagram below.



Which is true of the charges of the particles X and Y and the magnitude of the specific charge that is the ratio of charge to mass of particles X and Y?

	Charge of X	*Charge of Y*	*Magnitude of the specific charge*
A	Negative	Positive	$X < Y$
B	Negative	Positive	$X > Y$
C	Positive	Negative	$X < Y$
D	Positive	Negative	$X > Y$

Answer: B

With **B** into the page and both moving to the right, a positive charge is pushed upwards: Y is positive and X negative.

$$r = \frac{mv}{qB} \Longrightarrow \frac{q}{m} \propto \frac{1}{r}$$: X has the smaller radius, so the larger specific charge (B)

Explanation: Same velocity and same field.

12. Back e.m.f. is induced in a spinning electric motor. Which statement about the back e.m.f. is **not** true?

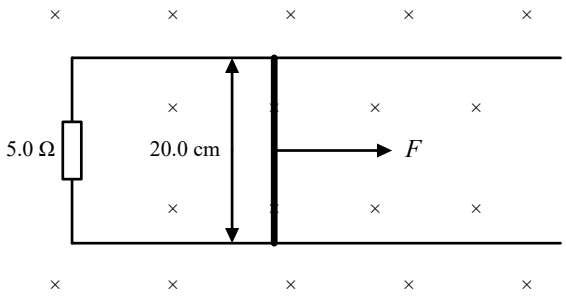
A It reduces the electric power supplied to the motor.
B Its magnitude is bigger when the motor starts to spin.
C Its direction is opposite to the direction of the voltage supplied.
D It is generated when an armature coil undergoes a magnetic flux change.

Answer: B

The back e.m.f. is proportional to the speed of rotation, so it is zero when the motor starts and largest at full speed. Statement B is not true.

Explanation: That is why a motor draws a large current when it starts.

13. A metal conductor of length 20.0 cm which is being moved with a constant velocity of 5.0 m s^{-1} by a constant force F in a uniform magnetic field of 4.0 T is shown in the diagram below.



The ends of the conducting track which has negligible resistance is connected to a resistor of 5.0 Ω. What is the magnitude of force F?

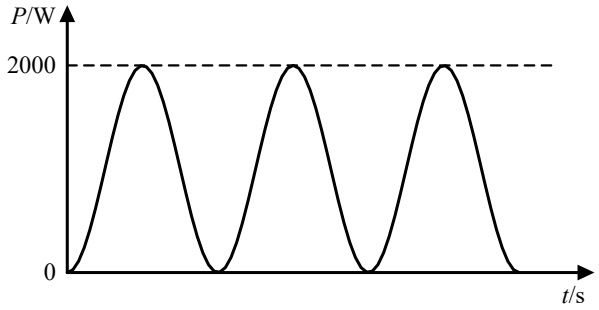
A 3.20 N B 0.80 N C 0.64 N D 0.16 N

Answer: C

$$F = \frac{B^2 l^2 v}{R} = \frac{(4.0)^2(0.200)^2(5.0)}{5.0} = 0.64 \text{ N}$$

Explanation: At constant velocity F balances the magnetic force BIl, with $I = Blv/R = 0.80$ A.

14. The variation of the instantaneous power P dissipated by a heating coil with time t when a sinusoidal current flows through the heating coil is shown in the graph below.



If the resistance of the coil is 100 Ω, what is the r.m.s. current flowing through the coil?

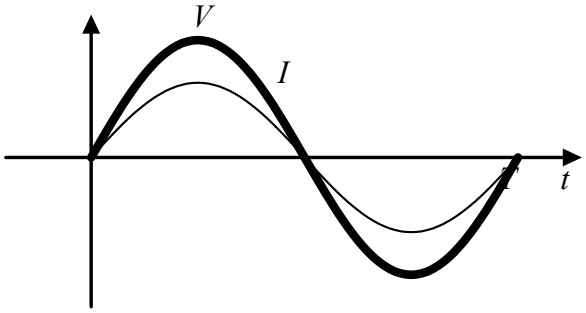
A 3.16 A B 4.47 A C 20.0 A D 28.3 A

Answer: A

$$\text{Mean power} = \frac{2000}{2} = 1000 \text{ W} = I_{\text{rms}}{}^2 R \Longrightarrow I_{\text{rms}} = \sqrt{\left(\frac{1000}{100}\right)} = 3.16 \text{ A}$$

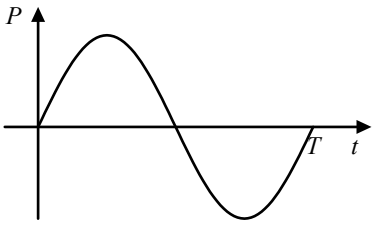
Explanation: The mean power is half the peak power for a resistor.

15. The variation of voltage V and current I with time t in a pure resistor is shown by the graph below.

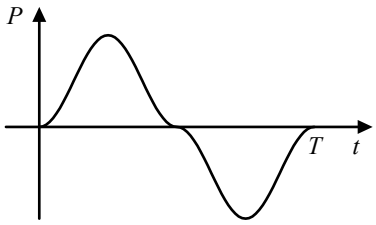


Which graph shows the variation of power P with time t?

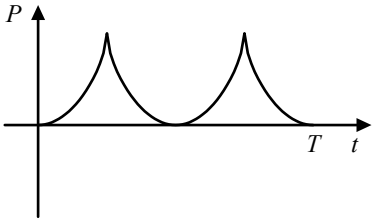
A



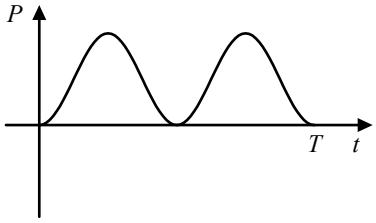
B



C



D



Answer: D

$P = VI \propto \sin^2 \omega t$: always positive, smoothly varying, with twice the frequency of V (D).

Explanation: In a resistor V and I are in phase.

Section B [15 marks]

16. (a) By using Gauss's law, derive an expression for an electric field which is due to a uniformly charged conducting plate of area A in a free space. The total charge on the plate is Q. [4]

Answer

Choose a cylindrical Gaussian surface (pillbox) of end area a crossing the surface of the plate, with one end inside the conductor	1
Inside the conductor $E = 0$, and no flux passes through the curved side; flux = Ea through the outer end	1
Gauss: $Ea = \frac{\sigma a}{\varepsilon_0} \Longrightarrow E = \frac{\sigma}{\varepsilon_0}$	1
With $\sigma = \frac{Q}{A}$: $E = \frac{Q}{A\varepsilon_0}$, perpendicular to the plate	1
	4

Explanation: If the charge is shared equally between both faces of a thin plate, $\sigma = Q/2A$ on each face and $E = Q/(2A\varepsilon_0)$ on each side.

(b) A charged particle of 40.0 μC is placed in a uniform electric field of 2.82×10^5 N C^{-1}. Determine

(i) the force exerted on the charged particle, [2]

Answer

$F = qE = (40.0 \times 10^{-6})(2.82 \times 10^5)$	1
$F = 11.3$ N	1
	2

Explanation: In the direction of the field for a positive charge.

(ii) the work which is required to move the charged particle to a distance 5.00 cm against the direction of the electric field. [2]

Answer

$W = Fd = (11.3)(0.0500)$ 1

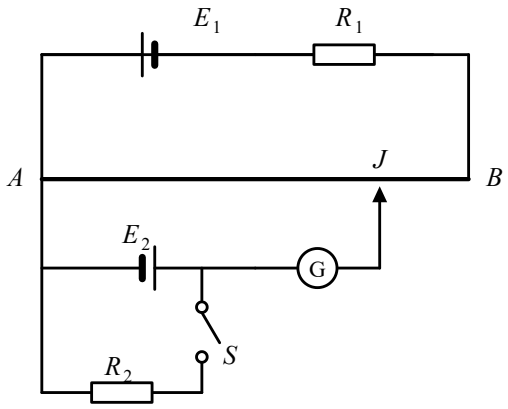
$W = 0.564$ J 1

2

Total for question **8**

Explanation: Work is done against the electric force.

17. A potentiometer is used to determine the e.m.f. and internal resistance of a cell E_2 as shown in the diagram below.



The battery E_1 has an e.m.f. of 2.0 V. It has a negligible internal resistance. The sliding wire AB is 100.0 cm long with a resistance of 4.0 Ω. The resistors R_1 and R_2 have resistance of 1.0 Ω and 5.0 Ω respectively. The balanced length of the wire is 85.0 cm when switch S is open, and it is 75.0 cm when the switch is closed.

(a) Determine the potential difference per unit length across the wire AB. [2]

Answer

$I = \frac{2.0}{1.0 + 4.0} = 0.40$ A; p.d. across $AB = (0.40)(4.0) = 1.6$ V 1

Potential gradient $= \frac{1.6}{1.000} = 1.6$ V m^{-1} 1

2

Explanation: R_1 is in series with the wire.

(b)

(i) Determine the e.m.f. for the cell E_2. [1]

Answer

$E_2 = (1.6)(0.850) = 1.36$ V — 1

1

Explanation: With S open no current flows through E_2 at balance.

(ii) Calculate the internal resistance of the cell E_2. [4]

Answer

S closed: terminal p.d. $V = (1.6)(0.750) = 1.20$ V — 1

$I = \frac{V}{R_2} = \frac{1.20}{5.0} = 0.24$ A — 1

$r = \frac{E_2 - V}{I} = \frac{1.36 - 1.20}{0.24}$ — 1

$r = 0.67\ \Omega$ — 1

4

Total for question **7**

Explanation: If E_2 is first rounded to 1.4 V, this gives 0.83 Ω.

Section C [30 marks]

18. (a) The expression for the electric conductivity σ is given by $\sigma = \frac{ne^2t}{m}$.

(i) State the meaning of the symbols n and t. [2]

Answer

n: the number of free charge carriers (electrons) per unit volume — 1

t: the mean time between collisions of the charge carriers with the lattice ions — 1

2

Explanation: e and m are the electron's charge and mass.

(ii) Use the above equation to explain the effect of temperature change on the resistivity of a semiconductor. [3]

Answer

In a semiconductor a rise in temperature frees many more charge carriers, so n increases greatly	1
t decreases slightly (more lattice vibration), but the increase in n dominates, so σ increases	1
Resistivity $\rho = \frac{1}{\sigma}$ decreases	1
	3

Explanation: This is the opposite of a metal.

(b) The relationship between current density J, electric conductivity σ and electric field strength E is given by $J = \sigma E$. Show that this equation is equivalent to Ohm's law. [4]

Answer

$J = \frac{I}{A}$ and $E = \frac{V}{l}$	1
$\frac{I}{A} = \sigma\frac{V}{l}$	1
$V = I(\frac{l}{\sigma A}) = I(\frac{\rho l}{A}) = IR$	1
For a given conductor at constant temperature, R is constant, so $V \propto I$: Ohm's law	1
	4

Explanation: $J = \sigma E$ is the microscopic form of Ohm's law.

(c) A wire has a diameter of 0.52 mm and a length of 40.0 cm. When a potential difference of 1.2 V is applied across both its ends, the current that flows is 2.5 A.

(i) Determine the resistivity of the wire. [2]

Answer

$R = \frac{1.2}{2.5} = 0.48\ \Omega$; $A = \pi(0.26 \times 10^{-3})^2 = 2.12 \times 10^{-7}\ \text{m}^2$	1
$\rho = \frac{RA}{l} = \frac{(0.48)(2.12 \times 10^{-7})}{0.400} = 2.55 \times 10^{-7}\ \Omega\ \text{m}$	1
	2

Explanation: Use the radius, half of 0.52 mm.

(ii) If the wire is cut into halves, what is the effect on its resistance and resistivity? Give a reason for your answer. [4]

Answer

The resistance is halved	1
because $R = \rho l/A \propto l$ and the length is halved (same area)	1
The resistivity is unchanged	1
because resistivity is a property of the material (at a given temperature), not of its dimensions	1
	4
Total for question	**15**

Explanation: Each half has resistance 0.24 Ω.

19. (a) A charged particle passing through a region of magnetic field moves in a straight line. State the direction of the magnetic field with respect to the motion of the particle. Explain your answer. [3]

Answer

The magnetic field is parallel (or antiparallel) to the velocity of the particle	1
The magnetic force $F = qvB \sin\theta$ is zero when $\theta = 0°$ or 180°	1
With no force acting, the particle continues in a straight line at constant velocity	1
	3

Explanation: Any perpendicular component of **v** would give a force.

(b) A proton moves in a circular path perpendicular to a constant magnetic field strength of 4.0×10^{-3} T. If the momentum of the proton is 1.52×10^{-23} kg m s^{-1}, determine

(i) the radius of the circular path, [4]

Answer

$evB = \frac{mv^2}{r}$	1
$r = \frac{mv}{eB} = \frac{p}{eB}$	1
$r = \frac{1.52 \times 10^{-23}}{(1.60 \times 10^{-19})(4.0 \times 10^{-3})}$	1
$r = 2.38 \times 10^{-2}$ m	1
	4

Explanation: The momentum is given directly.

(ii) the speed of the proton. [2]

Answer

$v = \frac{p}{m} = \frac{1.52 \times 10^{-23}}{1.67 \times 10^{-27}}$ 1

$v = 9.10 \times 10^{3}$ m s^{-1} 1

2

Explanation: Non-relativistic speed.

(c) A solenoid has 25 turns per cm, and carries a current of 5.00 mA. An electron inside the solenoid moves in a circular path with a speed of 1.50×10^{5} m s^{-1}.

(i) Determine the magnetic field strength in the solenoid. [2]

Answer

$n = 25$ turns per cm $= 2500$ m^{-1}; $B = \mu_0 nI = (4\pi \times 10^{-7})(2500)(5.00 \times 10^{-3})$ 1

$B = 1.57 \times 10^{-5}$ T 1

2

Explanation: The field inside a long solenoid is uniform.

(ii) Determine the magnetic force on the electron. State the reason why the force does not increase the kinetic energy of the electron. [2]

Answer

$F = evB = (1.60 \times 10^{-19})(1.50 \times 10^{5})(1.57 \times 10^{-5}) = 3.77 \times 10^{-19}$ N 1

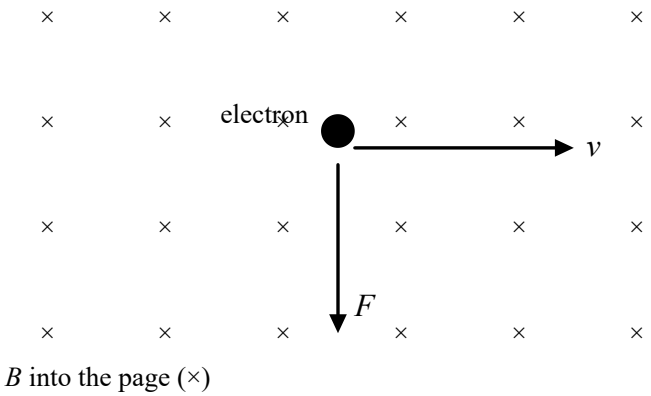
The force is always perpendicular to the velocity, so it does no work and the kinetic energy is unchanged 1

2

Explanation: It only changes the direction of motion.

(iii) Sketch a vector diagram to show the direction of the velocity ***v***, the magnetic field ***B*** and the magnetic force ***F*** for the electron. [2]

Answer



B into the page (×)

v, **B** and **F** mutually perpendicular	1
F in the direction opposite to that given by Fleming's left-hand rule, because the electron is negative	1
	2
Total for question	**15**

Explanation: $\mathbf{F} = -e\mathbf{v} \times \mathbf{B}$.

20. (a)

(i) State Faraday's law and Lenz's law of electromagnetic induction. [2]

Answer

Faraday: the induced e.m.f. is directly proportional to the rate of change of magnetic flux linkage	1
Lenz: the induced current flows in a direction that opposes the change producing it	1
	2

Explanation: Together they give the sign and size of the e.m.f.

(ii) Write an equation which represents these two laws. [1]

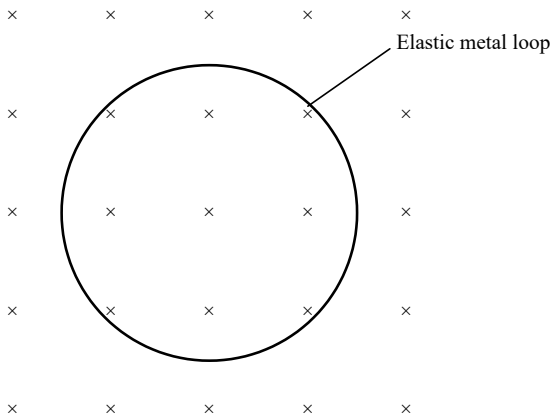
Answer

$\varepsilon = -\frac{\mathrm{d}(N\Phi)}{\mathrm{d}t}$ 1

1

Explanation: The minus sign expresses Lenz's law.

(b) An elastic metal loop which is stretched to a radius of 14 cm, and is placed in a uniform magnetic field is shown in the diagram below.



The magnetic flux density of 0.75 T passes perpendicularly through the plane of the loop. The loop suddenly shrinks to a radius of 9.0 cm in 0.16 s.

(i) Determine the induced e.m.f. in the loop. [3]

Answer

$\Delta A = \pi(0.14^2 - 0.090^2) = 3.61 \times 10^{-2}\ \text{m}^2$ 1

$\varepsilon = B\frac{\Delta A}{\Delta t} = (0.75)\frac{3.61 \times 10^{-2}}{0.16}$ 1

$\varepsilon = 0.17\ \text{V}$ 1

3

Explanation: The field is constant; the area changes.

(ii) State the direction of the induced current. [1]

Answer

Clockwise (as seen with the field into the page), to oppose the decrease of flux into the page 1

1

Explanation: Lenz's law.

(c) A d.c. motor draws an initial current of 1.2 A from a 3.0 V battery at the moment it is switched on. The current is 0.84 A when the armature of the motor spins at a speed of 45 revolutions per second. When this motor is loaded, it spins at 25 revolutions per second. Determine

(i) the resistance of the armature coil of the motor, [2]

Answer

At switch-on there is no back e.m.f.: $R = \frac{3.0}{1.2}$ 1

$R = 2.5\ \Omega$ 1

2

Explanation: The armature is not yet rotating.

(ii) the back e.m.f. produced when the motor spins at 45 revolutions per second, [2]

Answer

$\varepsilon_b = V - IR = 3.0 - (0.84)(2.5)$ 1

$\varepsilon_b = 0.90$ V 1

2

Explanation: The back e.m.f. reduces the current.

(iii) the output power of the motor at 45 revolutions per second, [2]

Answer

Output (mechanical) power $= \varepsilon_b I = (0.90)(0.84)$ 1

$= 0.756$ W 1

2

Explanation: The rest of the 2.52 W input is lost as heat in the armature (1.76 W).

(iv) the back e.m.f. at 25 revolution per second. [2]

Answer

Back e.m.f. $\propto$ speed: $\varepsilon = 0.90 \times \frac{25}{45}$ 1

$\varepsilon = 0.50$ V 1

2

Total for question **15**

Explanation: When loaded the motor slows and draws more current.